Near-Real-Time ISAC-Based Jammer Detection and Localization using srsRAN

Samanvitha Balijepalli
Dept. of ECE
Rice University
Houston, USA
sb273@rice.edu

Aniket Jadhav
Dept. of ECE
Rice University
Houston, USA
aj156@rice.edu

Sandra Mansour
Dept. of ECE
Rice University
Houston, USA
sm367@rice.edu

***Abstract*—Malicious jammers threaten wireless communications yet detecting them typically requires costly dedicated RF monitoring hardware. This work presents a software-based 5G framework for jammer detection and future localization using the srsRAN stack without hardware modification. The system reserves quiet symbol intervals within the 5G NR frame, creating standards-compliant sensing windows at only ≈1.4% scheduling overhead. SNR and jammer power extracted from these windows feed lightweight threshold-based classifiers. In a controlled ZMQ software testbed, jammer power achieves perfect classification (F1 = 1.0) while SNR-based detection reaches 92.18% accuracy (F1 = 0.957). A MUSIC-based angle-of-arrival baseline is separately validated, confirming readiness for Phase 2 localization. Although results reflect a simulated environment, they demonstrate feasible low-overhead jammer sensing and establish a concrete foundation for near-real-time localization on USRP hardware.**

***Index Terms*—Integrated Sensing and Communication (ISAC), jammer detection, jammer localization, srsRAN, 5G NR, wireless sensing, threshold-based detection, MUSIC, angle-of-arrival**

I. Introduction

Malicious radio-frequency jammers pose an increasingly significant threat to modern wireless systems. By flooding target bands with continuous-wave or broadband interference, they can disable GPS receivers, degrade LTE/5G base-station links, and disrupt safety-critical services such as emergency communications and unmanned aerial vehicle control [1]. Conventional jammer detection relies on dedicated spectrum-monitoring hardware — receivers, direction-finding antennas, and spectrum analysers — that must be separately purchased, maintained, and re-tuned whenever operating conditions change. This cost barrier motivates the development of software-defined alternatives that reuse existing communication infrastructure for sensing.

Integrated Sensing and Communication (ISAC) has emerged as a key direction for next-generation wireless systems, enabling communication infrastructure to simultaneously support sensing, detection, and localization tasks [2], [3]. By sharing radio resources between data transmission and environmental awareness, ISAC improves spectrum efficiency while eliminating the need for separate sensing hardware. Open and programmable 5G software stacks, such as srsRAN [4] and Open5GS [5], provide a particularly attractive experimental foundation because they expose scheduler internals that commercial equipment conceals.

Related Work

High-resolution subspace methods have long been the foundation of direction-of-arrival (DoA) estimation. MUSIC (Multiple Signal Classification), introduced by Schmidt [2], decomposes the array covariance matrix into signal and noise subspaces, yielding a pseudospectrum with sharp peaks at true source angles. Root-MUSIC [6] and ESPRIT [7] extend this framework by replacing grid search with polynomial rooting or subspace rotation, reducing computation while preserving resolution. Sparse reconstruction methods [8] offer further gains under limited snapshots but at higher algorithmic complexity. SpotFi [3] demonstrated that OFDM channel state information (CSI) — already collected during normal Wi-Fi communication — can feed a modified MUSIC pipeline for indoor localization, achieving sub-meter accuracy without additional hardware. A key limitation of SpotFi is that it assumes cooperative access points and a static, well-characterized indoor environment. Our work targets the more adversarial setting of jammer localization in a 5G NR environment, where the interference source is uncooperative, the signal structure is unknown, and software-defined gNB modifications are required to create sensing opportunities in the first place.

This work presents a software-based framework for jammer detection and future localization using srsRAN within an open 5G communication environment. Rather than relying on dedicated sensing signals or external RF instrumentation, the proposed system creates standards-compliant sensing opportunities directly from the communication process by reserving quiet symbol intervals within the 5G NR frame. As illustrated in Fig. 1, the current phase focuses on jammer detection using communication-quality indicators such as SNR and jammer power, while the future phase extends the framework toward localization using richer signal features and the MUSIC algorithm.

The main contributions of this work are: (i) a low-overhead, scheduler-based sensing design that creates quiet windows inside the 5G NR frame with only ≈1.4% resource overhead; (ii) an evaluation of threshold-based jammer detectors showing that jammer power achieves perfect separation in the current

dataset; and (iii) a validated MUSIC-based angle-of-arrival baseline that serves as the localization reference for the next phase. Together these results establish a practical foundation for near-real-time jammer localization in an open-source 5G platform.

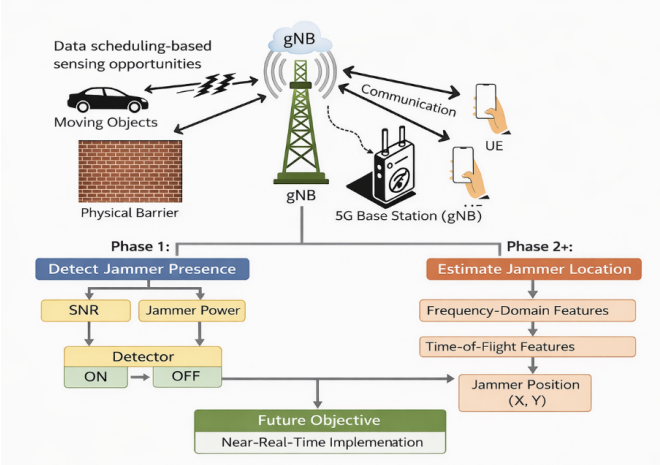


Fig. 1. Joint communication and sensing framework for jammer detection and localization using data-scheduling-based sensing opportunities in a 5G gNB. Phase 1 (current work) performs detection; Phase 2+ will estimate jammer position using frequency-domain and time-of-flight features fed into the MUSIC estimator.

II. Methodology

A. System Architecture

The proposed system is built on the srsRAN [4] 5G software stack and uses communication-driven sensing opportunities to detect and eventually localize jammer activity without dedicated sensing hardware. The full platform consists of an Open5GS [5] core network, a modified srsRAN gNB, an srsUE instance, and a jammer proxy inserted into the ZeroMQ (ZMQ) signal path, as shown in Fig. 2.

The gNB scheduler was modified to blank two OFDM symbols per sensing period, creating periodic *quiet windows* during which the transmitter is silent and the receiver captures ambient RF energy, interference, and jammer-induced signal components. IQ samples from these windows are stored for post-processing. Simultaneously, the jammer proxy logs UE power, jammer power, and signal-to-interference ratio (SIR) values synchronized to the same slot timeline, providing the ground-truth labels needed for detection evaluation.

A post-processing layer converts the captured IQ data and signal logs into MATLAB-compatible data structures. In the current stage, sensing is performed using communication-quality indicators derived from the scheduled windows — particularly SNR and jammer power — processed by lightweight threshold-based detectors. The longer-term goal is to extend this pipeline toward localization using richer signal features and the MUSIC algorithm [2], consistent with prior structured-measurement approaches such as SpotFi [3].

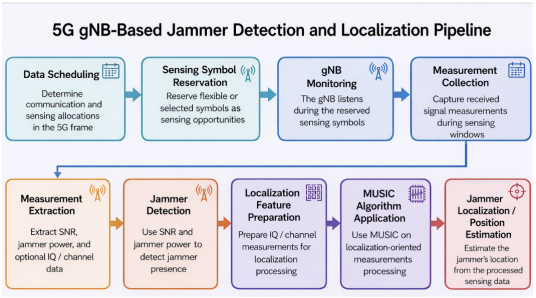


Fig. 2. System architecture and processing flow for the proposed sensing framework.

B. Data-Scheduling-Based Sensing Design

Rather than relying on Sounding Reference Signal (SRS) sweeps, the framework introduces sensing opportunities directly through data scheduling within the 5G NR frame, motivated by 3GPP TS 38.211 Table 4.3.2-1 [1], which designates symbols 12 and 13 as flexible symbols in standard TDD slot formats. When configured as downlink quiet symbols, these intervals let the gNB observe the channel without expecting UE transmission.

A new configuration structure was added to srsRAN to expose sensing-related parameters: enable flag, quiet symbol start index, quiet symbol count, and sensing period in slots. At 30 kHz subcarrier spacing one slot lasts 0.5 ms, and reserving 2 symbols once every 10 slots yields a sensing overhead of

$$\text{Overhead} = \frac{2}{10 \times 14} = \frac{2}{140} \approx 1.4\%.$$

This small resource cost creates periodic sensing opportunities within the normal communication process.

During each quiet sensing window, IQ samples are captured from the receive path and written to disk. The jammer proxy simultaneously injects controlled continuous-wave (CW) interference and logs UE power, jammer power, and SIR. These synchronized measurements are the input to the detection stage described next.

C. Detection and Localization Processing Framework

A MATLAB post-processing framework was developed to analyze the sensing measurements captured during the scheduled quiet windows. The main variables in the current stage are time, measured SNR, jammer power, and a ground-truth jammer-state label (OFF / ON).

To quantify class separability, summary statistics were computed. The sample mean and standard deviation are

$$\mu = \frac{1}{N}\sum_{i=1}^{N} x_i, \qquad \sigma = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N}(x_i - \mu)^2}. \tag{1}$$

Using class means, the midpoint thresholds are

$$T_{\text{SNR}} = \frac{\mu_{\text{SNR,OFF}} + \mu_{\text{SNR,ON}}}{2}, \quad T_{\text{JP}} = \frac{\mu_{\text{JP,OFF}} + \mu_{\text{JP,ON}}}{2}. \tag{2}$$

The three detectors are defined as

$$\hat{y}_{\mathrm{SNR}}(i) = \begin{cases} 1, & \mathrm{SNR}(i) < T_{\mathrm{SNR}} \\ 0, & \text{otherwise,} \end{cases} \tag{3}$$

$$\hat{y}_{\mathrm{JP}}(i) = \begin{cases} 1, & P_J(i) > T_{\mathrm{JP}} \\ 0, & \text{otherwise,} \end{cases} \tag{4}$$

$$\hat{y}_{\mathrm{C}}(i) = \begin{cases} 1, & \hat{y}_{\mathrm{SNR}}(i) = 1 \ \vee \ \hat{y}_{\mathrm{JP}}(i) = 1 \\ 0, & \text{otherwise.} \end{cases} \tag{5}$$

Overall accuracy is

$$\mathrm{Acc} = \frac{TP + TN}{TP + TN + FP + FN}. \tag{6}$$

Because the dataset is class-imbalanced (501 OFF vs. 8953 ON samples, roughly a 1:18 ratio), accuracy alone can be misleading. We therefore also report precision $P = TP/(TP + FP)$, recall $R = TP/(TP + FN)$, and F1-score $F_1 = 2PR/(P + R)$ for each detector.

To support the future localization stage, a separate framework evaluates MUSIC-based angle-of-arrival (AoA) estimation using a 3-element uniform linear array (ULA). Spatial covariance matrix decomposition separates the signal and noise subspaces, after which the MUSIC pseudospectrum is evaluated for angle estimation. This stage is not yet integrated into the srsRAN sensing path but provides a validated reference model consistent with prior high-resolution methods [2], [6]–[8].

D. srsRAN Test Environment

The full system was deployed on a Linux server using the srsRAN 5G software stack, Open5GS core network, and ZMQ-based simulated radio links. The system was configured for NR Band 3 (center frequency 1842.5 MHz, 20 MHz bandwidth, 30 kHz subcarrier spacing, single-antenna TDD, single-user scenario).

The gNB and UE communicated normally during standard operation. The jammer proxy introduced a configurable CW interference signal after slot 2000, injecting at 100 kHz frequency offset with amplitude 0.8. Runtime execution was monitored via command-line logs to verify gNB initialization, UE attachment, and jammer activation.

Testbed limitation: ZMQ-based transport introduces padding values into the receive buffer during certain quiet-symbol intervals. Some captured IQ samples therefore reflect simulation transport behavior rather than true ambient RF. This does not invalidate the threshold-based detection results — which rely on the jammer proxy's synchronized log values rather than raw IQ power — but it does mean the present results should be interpreted as a controlled software testbed study. A future USRP hardware implementation would capture actual ambient RF and jammer energy directly.

III. Results

A. Jammer Injection and Measurement Overview

Experiments were conducted to evaluate jammer detection using the proposed scheduler-based sensing pipeline. A total of 9454 sensing measurements were recorded: 501 jammer OFF samples (slots 0–1999) and 8953 jammer ON samples (slots 2000 onward), as summarized in Table I.

Before jammer activation the average measured SNR was approximately 11.07 dB. After activation the SNR dropped sharply to approximately -91.10 dB, indicating severe communication degradation. Simultaneously, measured jammer power increased from 0 to approximately 0.64. These feature shifts establish a clear distinction between the two states and motivate the threshold-based detection results presented below.

TABLE I
DATASET STATISTICS UNDER JAMMER OFF AND JAMMER ON CONDITIONS.

Metric	Value
Total samples	9454
Jammer OFF samples	501
Jammer ON samples	8953
Class ratio (OFF:ON)	1:18
SNR mean (OFF)	11.07 dB
SNR std. (OFF)	31.23 dB
SNR mean (ON)	-91.10 dB
SNR std. (ON)	26.34 dB
Jammer power mean (OFF)	0.00
Jammer power std. (OFF)	0.00
Jammer power mean (ON)	0.64
Jammer power std. (ON)	0.00

B. Feature Behavior Over Time

Fig. 3 shows jammer power over time, split by jammer state. The OFF samples cluster near zero; the ON samples cluster near 0.64 with no visible overlap. Fig. 4 shows the measured SNR over time. During normal operation SNR is high and positive; after jammer activation it collapses to strongly negative values, with intermittent spikes caused by repeated UE reconnection attempts.

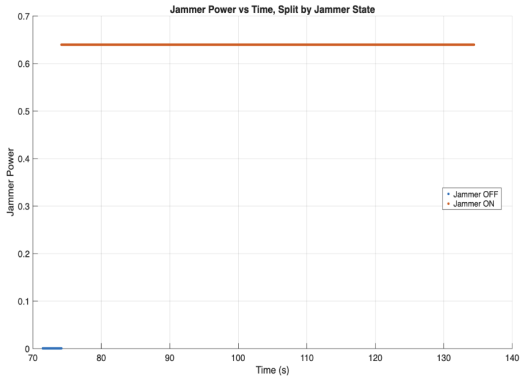


Fig. 3. Jammer power versus time for jammer OFF and jammer ON conditions.

C. Threshold-Based Jammer Detection

Three threshold-based detectors were evaluated; numerical performance is summarized in Table II.

Fig. 5 shows the SNR-based detector. The jammer OFF and ON SNR distributions are substantially separated, but

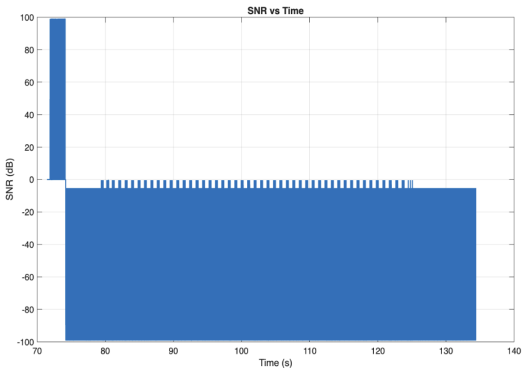


Fig. 4. SNR versus time under jammer sensing conditions.

TABLE II
PERFORMANCE OF THRESHOLD-BASED JAMMER DETECTORS.

Detector	Thresh.	Acc.	Prec.	Rec.	F1	TP	TN	FP	FN
SNR-based	−40.02 dB	92.18%	100%	91.74%	0.957	8214	501	0	739
Jammer-power	0.320	**100%**	100%	100%	1.000	8953	501	0	0
Combined	N/A	**100%**	100%	100%	1.000	8953	501	0	0

some spread near the threshold leaves the classes imperfectly separable. The detector achieved 92.18% accuracy with 739 false negatives and an F1-score of 0.957. The false negatives correspond to intermittent SNR spikes during UE reconnection attempts, which temporarily push jammer-ON measurements above the threshold.

Fig. 6 shows the jammer-power-based detector. The OFF and ON distributions are completely disjoint: all OFF values sit at exactly 0 and all ON values sit at 0.64. The detector achieved 100% accuracy with F1 = 1.0 and no errors, providing the strongest first-stage detection feature.

The combined detector (Eq. (5)) also achieves 100% accuracy and F1 = 1.0, but does not improve on the jammer-power detector alone. This is expected: jammer power already provides perfect separation in the current dataset, so adding the SNR criterion contributes nothing extra. Its value would likely emerge under more challenging conditions (see Section IV).

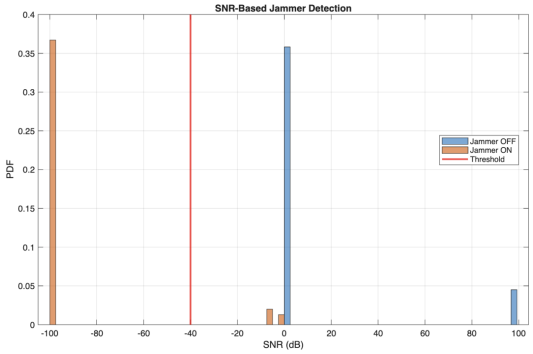


Fig. 5. SNR-based jammer detection histogram. The vertical red line marks the decision threshold at −40.02 dB.

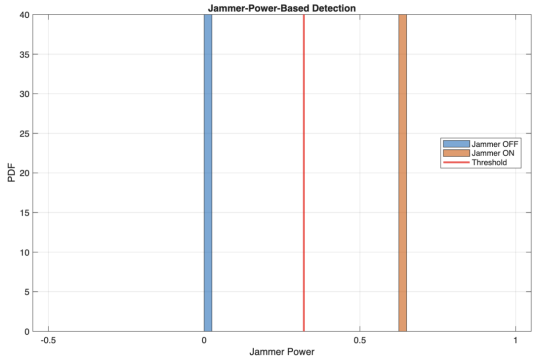


Fig. 6. Jammer-power-based detection histogram. The two classes are completely separated with no overlap.

D. MUSIC Localization Baseline

To support the future localization stage, a separate MATLAB simulation validated a MUSIC-based AoA baseline using a 3-element ULA. Fig. 7 presents four diagnostics:

- **CSI phase:** Approximately stable across subcarriers for all three antennas, indicating a well-conditioned input with preserved spatial phase structure.
- **I-Q constellation:** Tight clustering confirms acceptable baseband signal integrity.
- **Covariance eigenvalues:** One dominant eigenvalue, consistent with a rank-1 single-source assumption.
- **MUSIC pseudospectrum:** Clear dominant peak aligned with the 25° ground-truth angle.

Together these results confirm that the MUSIC estimator correctly isolates the signal subspace [2] and provides a valid localization baseline for Phase 2 integration.

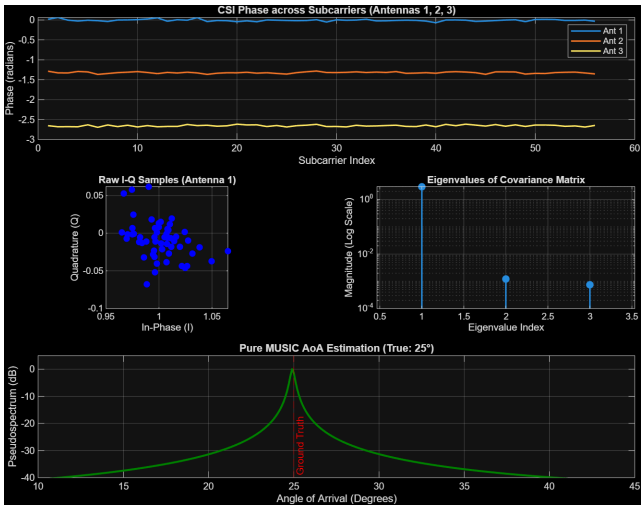


Fig. 7. MUSIC-based angle-of-arrival estimation baseline: CSI phase (top-left), I-Q constellation (bottom-left), covariance eigenvalues (top-right), and MUSIC pseudospectrum with peak at 25° (bottom-right).

IV. Discussion

Detection performance. Jammer activity produces strong, consistent feature changes: mean SNR drops from 11.07 dB to −91.10 dB and mean jammer power rises from 0 to 0.64 (Table I). The jammer-power detector achieves perfect classification (F1 = 1.0), while the SNR detector achieves F1 = 0.957. The 739 SNR false negatives arise from UE reconnection spikes that temporarily elevate measured SNR during the jammer-ON period, reflecting the greater variability of SNR as a communication-quality metric.

Class imbalance. The dataset contains roughly 18× more jammer-ON than jammer-OFF samples (Table I), a direct consequence of the experimental design: jammer activation was delayed to slot 2000 to establish a clean baseline, leaving a short OFF window. Because the jammer-power detector has zero FP and zero FN, the imbalance does not affect its evaluation; all metrics equal 1.0 regardless of class weights. The SNR detector is similarly unaffected by imbalance in the FP direction (zero FP), and precision remains 100%. Future experiments should balance the class distribution more carefully or apply stratified cross-validation to ensure that results generalize across different ON/OFF ratios.

Testbed limitations. As noted in Section II-D, the ZMQ transport introduces IQ padding artifacts during quiet-symbol intervals. The present detection results are not invalidated by this — they rely on the proxy log values rather than raw IQ power — but a hardware deployment on USRP radios is required before the IQ-domain features needed for MUSIC-based localization can be captured reliably from actual RF.

Interpreting the 100% accuracy result. Perfect classification in a detection experiment warrants scrutiny. In the current setup, jammer power is logged directly by the proxy — it reflects the injected signal amplitude rather than an estimated RF quantity derived from IQ data. The separation is therefore exact by construction: OFF samples carry exactly 0 power and ON samples carry exactly 0.64. This result demonstrates that the *measurement pipeline* reliably captures jammer state, but it does not yet establish robustness against realistic impairments such as multipath fading, antenna pattern variation, background interference, or jammers that modulate their power over time. SNR, with its 92.18% accuracy, is a better proxy for true RF-domain difficulty.

Combined detector. The combined detector matches jammer-power performance exactly, confirming that jammer power dominates in the current experiment. This does not imply SNR is irrelevant: in a hardware deployment with fading or when the jammer amplitude varies, SNR may provide complementary evidence that improves robustness. A weighted fusion or machine-learning classifier could exploit both features more flexibly in future work.

MUSIC baseline and Phase 2 roadmap. The stable CSI phase, tight I-Q clustering, dominant covariance eigenvalue, and 25°-aligned pseudospectrum peak confirm that the MUSIC estimator is correctly isolating the signal subspace [2]. The planned Phase 2 integration proceeds in three steps: (i) port IQ capture from ZMQ to USRP hardware; (ii) extract per-subcarrier CSI matrices from the quiet-window samples; and (iii) feed a multi-antenna MUSIC pipeline with those CSI matrices to estimate jammer AoA in near real time. Combining AoA from two or more gNBs will then yield a 2-D jammer position estimate.

Summary. Communication-driven sensing opportunities created through scheduling are sufficient for reliable first-stage jammer detection in an open-source 5G framework. Jammer power is the most discriminative feature under the present conditions, while SNR remains a useful secondary feature whose value will likely grow in more challenging RF environments. These findings validate the low-overhead sensing architecture and establish a concrete roadmap for extending the system to near-real-time jammer localization.

The entire source code and related images can be found at this GitHub repository.

Acknowledgments

This work was conducted as part of the Master of Electrical and Computer Engineering program at Rice University. The authors acknowledge the academic support and institutional resources provided by Rice University, as well as the guidance and feedback from Dr. Rahman Doost throughout the project. The statements made in this work are solely the responsibility of the authors.

CRediT Author Statement

Sandra Mansour: Software, Formal analysis, Visualization, Writing—Original draft. **Samanvitha Balijepalli:** Investigation, Algorithm analysis, Writing—Review & Editing. **Aniket Jadhav:** Investigation, Methodology, Software. **Dr. Rahman Doost:** Conceptualization, Supervision, Project administration, Resources, Writing—Review & Editing.

References

[1] 3GPP, *NR; Physical channels and modulation*, 3rd Generation Partnership Project (3GPP) Std. TS 38.211, 2024, version 18 series.

[2] R. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Transactions on Antennas and Propagation*, vol. 34, no. 3, pp. 276–280, 1986.

[3] M. Kotaru, K. Joshi, D. Bharadia, and S. Katti, "Spotfi," *ACM SIGCOMM Computer Communication Review*, vol. 45, pp. 269–282, 08 2015.

[4] Software Radio Systems. (2024) srsRAN Project Documentation. [Online; accessed 14-March-2024]. [Online]. Available: https://docs.srsran.com/projects/project/en/latest/

[5] Open5GS Authors. (2025) Open5GS: C-language Open Source implementation of 5G Core and EPC. [Online; accessed 14-Mar-2025]. [Online]. Available: https://open5gs.org/

[6] B. Rao and K. Hari, "Performance analysis of root-music," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 37, no. 12, pp. 1939–1949, 1989.

[7] R. Roy and T. Kailath, "Esprit-estimation of signal parameters via rotational invariance techniques," *IEEE Transactions on Acoustics, Speech, and Signal Processing*, vol. 37, no. 7, pp. 984–995, 1989.

[8] Z. Yang, J. Li, P. Stoica, and L. Xie, *Sparse methods for direction-of-arrival estimation*, 01 2018, pp. 509–581.